CSE 471
Summer 2026
Assignment #2 (10 marks)
Deadline: 21 July 2026 by 11:59 PM

Instructions for preparing the solution script:

- **Try to complete the assignment in one page (Your name, id, section in the same page on top)**
- **You have to submit a handwritten scanned copy of your assignment.**
- **Late submissions will not be accepted.**

A courier service system named Mangrove allows users to send, track, and receive parcels through a mobile application.
The User submits a delivery request via the MobileApp with receiver details and parcel information. The MobileApp sends the request to the CourierSystem, which validates the input. If the input is invalid, the CourierSystem rejects the request and the app asks the User to re-enter information. If the input is valid, the CourierSystem creates a DeliveryOrder and forwards the parcel to the DeliveryHub by a CourierAgent. For this, the CourierSystem assigns a CourierAgent to pick up the parcel. The CourierAgent collects the parcel from the User, forwards it and updates its status in the TrackingSystem. A TrackingID is generated and sent to both User and Receiver. During transit, the parcel status (e.g., “In Transit,” “Arrived at Hub,” “Out for Delivery”) is continuously updated. The User or Receiver can request real time tracking via the MobileApp. At the destination, a DeliveryAgent is assigned by the CourierSystem to deliver the parcel. The DeliveryAgent attempts delivery and requests confirmation from the Receiver. If the delivery is successful, the DeliveryAgent updates the TrackingSystem to “Delivered,” both User and Receiver are notified through App, and the order is closed. If the receiver is unavailable, the DeliveryAgent marks a failed attempt, and the CourierSystem reschedules delivery (up to two times). If all attempts fail, the parcel is returned to the User. If the receiver cancels before delivery via the MobileApp, the CourierSystem stops the delivery, and the parcel is returned to the User. Finally, the system updates the final status in the TrackingSystem and terminates the delivery process.

[10 marks] Design a Sequence Diagram based on the above scenario.